

CUSTOMER CHURN ANALYSIS

An End-to-End Data Analytics Project

Python → MySQL → SQL → Power BI

Project Report

Dataset: 521 Customers

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1. Project Overview

Customer churn is an important business problem because losing existing customers can affect customer retention and revenue. This project analyzes customer churn using customer demographic information, subscription details, tenure, revenue-related information, and customer support interactions.

The project follows an end-to-end data-analysis workflow:

Excel → Python → MySQL → SQL → Power BI

Python was used for data cleaning, preparation, feature engineering, exploratory data analysis, and visualization. MySQL was used to store the prepared dataset and perform SQL-based business analysis. Power BI was used to create an interactive five-page dashboard for presenting the major findings.

The final analytical dataset contains 521 customers and includes customer, subscription, churn, tenure, revenue, risk, complaint, escalation, and customer satisfaction information.

2. Business Problem

The objective of this project is to understand customer churn patterns and identify customer segments or characteristics associated with observed churn.

- What percentage of customers have churned?
- How does churn vary across customer groups?
- Does customer tenure differ between churned and active customers?
- Does subscription, plan, or contract type vary in churn rate?
- How does churn relate to revenue?
- Do customers with complaints have different observed churn rates?
- How does customer satisfaction differ between churned and active customers?
- Which states have the highest numbers of churned customers?

The project focuses on descriptive analysis of the available dataset rather than building a predictive churn model.

3. Project Objectives

1. Clean and prepare the raw customer datasets.
2. Combine customer, subscription, and support information.
3. Create useful analytical features.
4. Analyze customer churn using Python.
5. Perform business-oriented SQL analysis using MySQL.
6. Build an interactive Power BI dashboard.
7. Identify important patterns related to customer churn.
8. Present the findings in a clear and business-oriented format.

4. Dataset Description

4.1 Customer Data

- Customer ID
- Customer name
- Country

- State
- Gender
- Date of birth
- Age
- Age group

4.2 Subscription Data

- Customer ID
- Subscription start date
- Subscription type
- Renewal date
- Plan type
- Contract type
- Cancellation date
- Cancellation reason
- Monthly charges
- CLTV
- Churn score
- Churn flag
- Tenure
- Annual charges
- Tenure group
- Churn risk group

4.3 Customer Support Data

- Customer ID
- Complaint date
- Escalation status
- CSAT score
- Complaint count
- Escalation count
- Average CSAT score
- Last complaint date

After cleaning and combining the datasets, the final analytical dataset contained 521 customer records.

5. Data Cleaning and Preparation

Python and pandas were used to prepare the datasets for analysis.

- Checking dataset structure and data types.
- Checking missing values.
- Removing unnecessary columns.
- Converting date columns into appropriate datetime formats.
- Checking duplicate records.
- Handling missing country information using available state-country relationships.
- Standardizing relevant fields.
- Checking customer-level uniqueness.

- Aggregating customer support records where multiple support records existed for the same customer.
- Combining the customer, subscription, and support datasets.

The customer and subscription datasets contained one record per customer, while the support dataset could contain multiple records for the same customer. Therefore, support information was aggregated at the customer level before merging.

The final dataset was used for Python analysis, SQL analysis, and Power BI reporting.

6. Feature Engineering

Age

Age was calculated from the customer's date of birth.

Age Group

Customers were grouped into 18–25, 26–35, 36–45, 46–55, and 56–65.

Churn Flag

A binary churn indicator was created: 0 = Not Churned and 1 = Churned. The flag was derived from cancellation information.

Tenure

Customer tenure was calculated using subscription dates and represented in days and years.

Tenure Group

Customers were grouped into 0–1 years, 1–3 years, 3–5 years, 5–7 years, and 7+ years.

Annual Charges

Annual charges were calculated as $\text{Monthly Charges} \times 12$.

Churn Risk Group

Customers were grouped according to their churn scores into risk categories.

Support Features

Support-related features included escalation flag, complaint year/month, complaint count, escalation count, average CSAT score, and last complaint date.

7. Exploratory Data Analysis

Exploratory analysis was performed using Python to understand customer characteristics and identify patterns in churn.

- Age
- Gender
- Geography
- Subscription type
- Plan type
- Contract type

- Tenure
- Revenue
- Churn risk
- Complaints
- Escalations
- Customer satisfaction

A major pattern observed in the analysis was the difference in churn across customer-tenure groups. Shorter-tenure customers showed substantially higher observed churn than longer-tenure customers in this dataset.

The analysis also showed differences in churn across plans, subscription types, age groups, and geographic areas.

8. SQL Analysis

The prepared dataset was loaded into MySQL.

Database: customer_churn_analysis

Table: customer_churn_data

Q1. Total Number of Customers

```
mysql> SELECT COUNT(DISTINCT customerid) AS total_customers_number FROM customer_churn_data;
+-----+
| total_customers_number |
+-----+
|                521 |
+-----+
1 row in set (0.01 sec)
```

The dataset contains 521 customers.

Q2. Overall Churn Rate

```
mysql> SELECT ROUND(SUM(churn_flag) * 100.0 / COUNT(*), 2) AS churn_rate FROM customer_churn_data;
+-----+
| churn_rate |
+-----+
|      26.30 |
+-----+
1 row in set (0.09 sec)
```

The overall observed churn rate is 26.30%.

Q3. Customers by Contract Type

```
mysql> SELECT contract_type, COUNT(customerid) AS churned_customer FROM customer_churn_data GROUP BY contract_type;
+-----+-----+
| contract_type | churned_customer |
+-----+-----+
| Annual       |                254 |
| Monthly      |                267 |
+-----+-----+
2 rows in set (0.04 sec)
```

Contract Type	Customers
Annual	254
Monthly	267

Q4. Churn Rate by Subscription Type

```
mysql> SELECT subscription_type, ROUND(SUM(churn_flag) * 100 / COUNT(*), 2) AS churn_rate FROM customer_churn_data GROUP BY subscription_type;
```

subscription_type	churn_rate
Referral	29.35
Paid	23.64
Organic	25.58

Subscription Type	Churn Rate
Referral	29.35%
Paid	23.64%
Organic	25.58%

Q5. Average Tenure of Churned vs Active Customers

```
mysql> SELECT CASE
  -> WHEN churn_flag = 0 THEN 'Not Churned'
  -> WHEN churn_flag = 1 THEN 'Churned'
  -> END as churn_status,
  -> ROUND(AVG(tenure_years), 2) AS average_tenure
  -> FROM customer_churn_data GROUP BY churn_flag;
```

churn_status	average_tenure
Not Churned	4.64
Churned	1.68

2 rows in set (0.03 sec)

Customer Status	Average Tenure
Not Churned	4.64 years
Churned	1.68 years

Q6. Annualized Revenue of Active vs Churned Customers

```
mysql> SELECT CASE
  -> WHEN churn_flag = 0 THEN 'Not Churned'
  -> When churn_flag = 1 THEN 'Churned'
  -> END AS churn_status,
  -> ROUND(SUM(monthly_charges) * 12, 2) AS annual_revenue
  -> FROM customer_churn_data GROUP BY churn_flag;
```

churn_status	annual_revenue
Not Churned	68831.04
Churned	24872.64

2 rows in set (0.01 sec)

Customer Status	Annualized Revenue
Not Churned	68,831.04
Churned	24,872.64

Q7. Churn Rate by Plan Type

```
mysql> SELECT plan_type, ROUND(SUM(churn_flag) * 100 / COUNT(*), 2) AS churn_rate FROM customer_churn_data GROUP BY plan_type;
+-----+-----+
| plan_type | churn_rate |
+-----+-----+
| Standard | 26.60 |
| Premium | 28.91 |
| Basic | 24.21 |
+-----+-----+
3 rows in set (0.02 sec)
```

Plan Type	Churn Rate
Standard	26.60%
Premium	28.91%
Basic	24.21%

Q8. Churn Rate by Age Group

```
mysql> SELECT age_group, ROUND(SUM(churn_flag)*100 / COUNT(*), 2) AS churn_rate FROM customer_churn_data GROUP BY age_group;
+-----+-----+
| age_group | churn_rate |
+-----+-----+
| 36-45 | 26.53 |
| 26-35 | 30.28 |
| 46-55 | 24.11 |
| 18-25 | 23.19 |
| 56-65 | 26.19 |
+-----+-----+
5 rows in set (0.01 sec)
```

Age Group	Churn Rate
36-45	26.53%
26-35	30.28%
46-55	24.11%
18-25	23.19%
56-65	26.19%

Q9. Churn Rate by Tenure Group

```
mysql> SELECT
-> tenure_group,
-> ROUND(SUM(churn_flag) * 100.0 / COUNT(*), 2) AS churn_rate
-> FROM customer_churn_data
-> GROUP BY tenure_group
-> ORDER BY churn_rate DESC;
+-----+-----+
| tenure_group | churn_rate |
+-----+-----+
| 0-1 years | 100.00 |
| 1-3 years | 41.50 |
| 3-5 years | 12.50 |
| 5-7 years | 3.73 |
| 7+ years | 0.00 |
+-----+-----+
5 rows in set (0.02 sec)
```

Tenure Group	Churn Rate
0-1 years	100.00%
1-3 years	41.50%
3-5 years	12.50%
5-7 years	3.73%
7+ years	0.00%

Q10. Churn Rate: Customers With vs Without Complaints

```
mysql> SELECT CASE
  -> WHEN complaint_count > 0 THEN 'With Complaints'
  -> ELSE 'Without Complaints'
  -> END AS complaint_status,
  -> COUNT(*) AS total_customers,
  -> SUM(churn_flag) AS churned_customers,
  -> ROUND(SUM(churn_flag) * 100.0 / COUNT(*), 2) AS churn_rate
  -> FROM customer_churn_data GROUP BY complaint_status;
+-----+-----+-----+-----+
| complaint_status | total_customers | churned_customers | churn_rate |
+-----+-----+-----+-----+
| Without Complaints | 339 | 80 | 23.60 |
| With Complaints | 182 | 57 | 31.32 |
+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

Complaint Status	Customers	Churned	Churn Rate
Without Complaints	339	80	23.60%
With Complaints	182	57	31.32%

Customers with recorded complaints had a higher observed churn rate in this dataset. This is an observed association and does not by itself establish that complaints caused customers to churn.

Q11. Average CSAT Score by Churn Status

```
mysql> SELECT CASE
  -> WHEN churn_flag = 1 THEN 'Churned'
  -> WHEN churn_flag = 0 THEN 'Active'
  -> END AS customer_status,
  -> COUNT(*) AS total_customers,
  -> ROUND(AVG(avg_csat_score), 2) AS average_csat_score
  -> FROM customer_churn_data
  -> WHERE avg_csat_score IS NOT NULL GROUP BY churn_flag;
+-----+-----+-----+
| customer_status | total_customers | average_csat_score |
+-----+-----+-----+
| Churned | 57 | 38.46 |
| Active | 125 | 35.98 |
+-----+-----+-----+
2 rows in set (0.01 sec)
```

Among customers with available CSAT information:

Customer Status	Customers	Average CSAT
Churned	57	38.46
Active	125	35.98

Q12. Churned Customers by State

```
mysql> SELECT state,
  -> COUNT(*) AS total_customers,
  -> SUM(churn_flag) AS churned_customers,
  -> ROUND(SUM(churn_flag) * 100.0 / COUNT(*), 2) AS churn_rate
  -> FROM customer_churn_data
  -> GROUP BY state ORDER BY churned_customers DESC;
```

state	total_customers	churned_customers	churn_rate
Kathmandu	59	19	32.20
Delhi	63	15	23.81
Telangana	44	15	34.09
Kerala	34	13	38.24
Karnataka	32	10	31.25
Uttar Pradesh	39	10	25.64
Meghalaya	24	9	37.50
Tamil Nadu	24	7	29.17
Punjab	27	6	22.22
Gujarat	23	6	26.09
Maharashtra	22	5	22.73
Nagaland	25	5	20.00
West Bengal	24	5	20.83
Bihar	31	5	16.13
Rajasthan	31	4	12.90
Odisha	19	3	15.79

16 rows in set (0.01 sec)

State	Customers	Churned	Churn Rate
Kathmandu	59	19	32.20%
Delhi	63	15	23.81%
Telangana	44	15	34.09%
Kerala	34	13	38.24%
Karnataka	32	10	31.25%
Uttar Pradesh	39	10	25.64%
Meghalaya	24	9	37.50%
Tamil Nadu	24	7	29.17%
Punjab	27	6	22.22%
Gujarat	23	6	26.09%
Maharashtra	22	5	22.73%
Nagaland	25	5	20.00%
West Bengal	24	5	20.83%
Bihar	31	5	16.13%
Rajasthan	31	4	12.90%
Odisha	19	3	15.79%

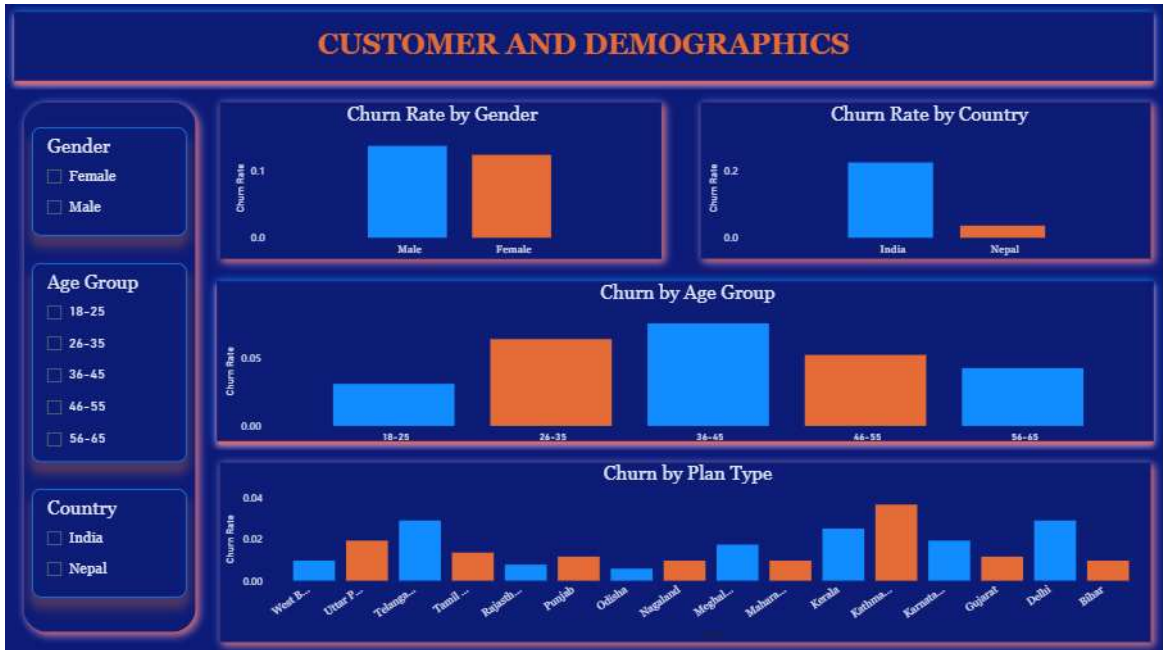
9. Power BI Dashboard

An interactive Power BI dashboard was developed using the prepared customer churn dataset. The dashboard contains five analytical pages.

Page 1 — Churn Overview



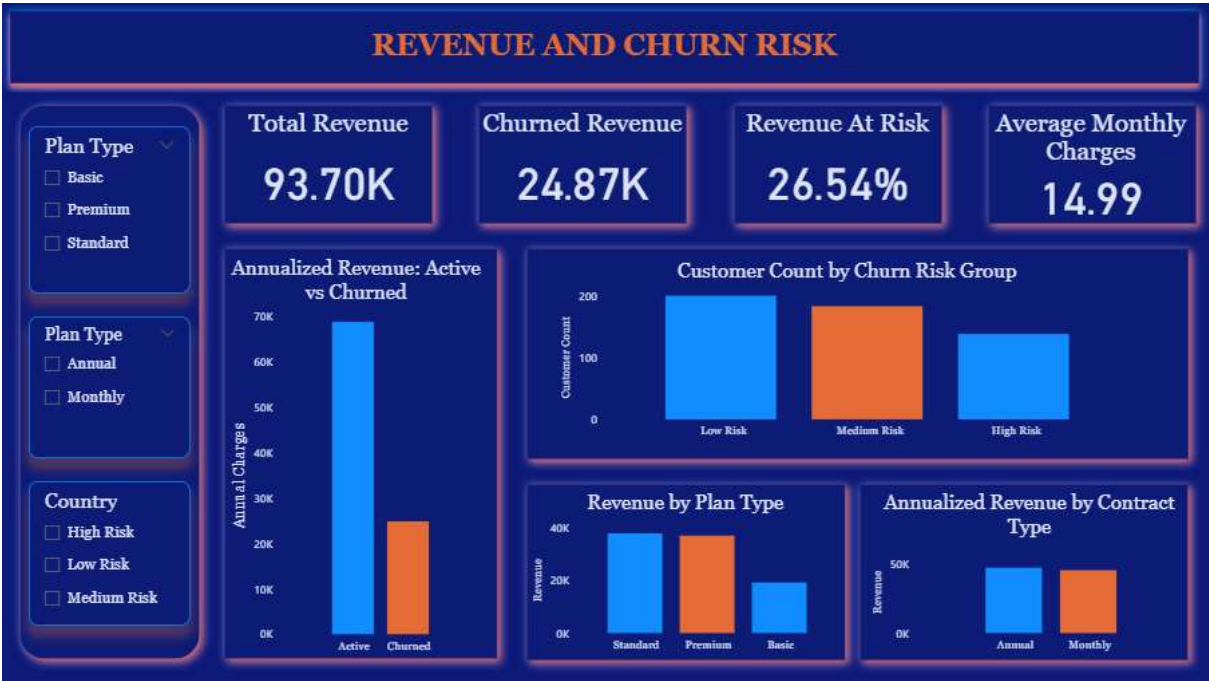
Page 2 — Customer Demographics

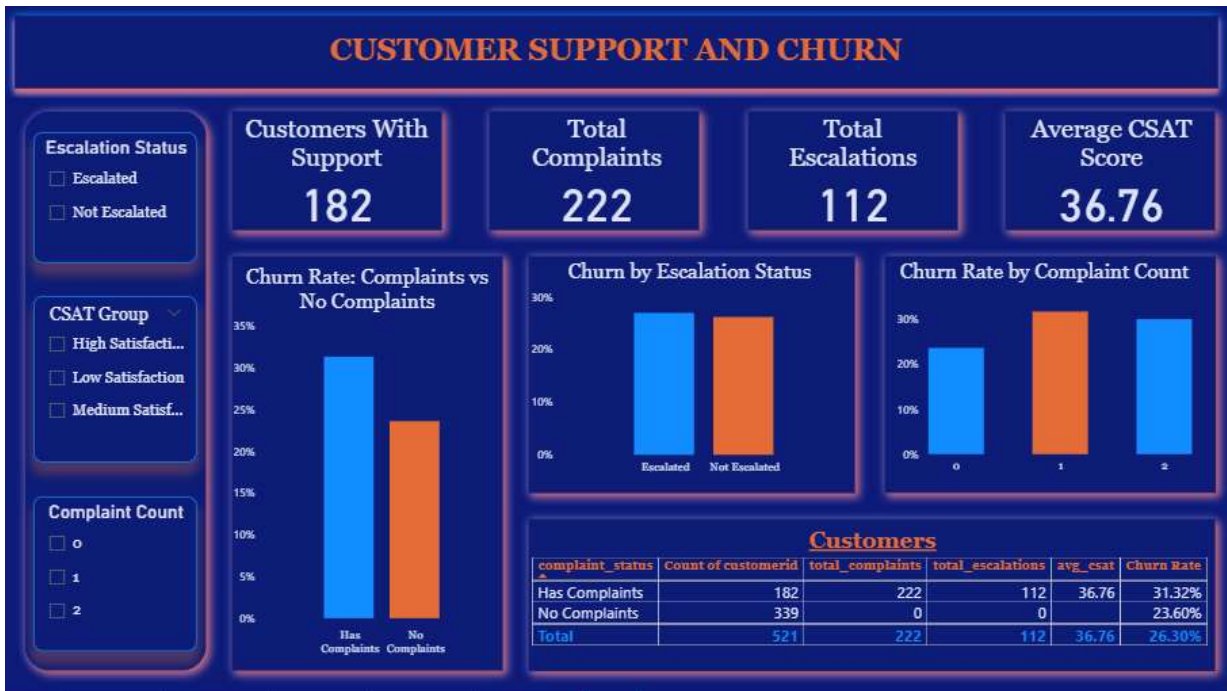


Page 3 — Subscription and Tenure



Page 4 — Revenue and Churn Risk





10. Key Findings

Overall churn: The dataset contains 521 customers, with an observed churn rate of 26.30%.

Tenure and churn: Churn varies substantially across tenure groups. The 0–1 year group had an observed churn rate of 100.00%, while the 7+ year group had 0.00%. Average tenure was 1.68 years for churned customers and 4.64 years for customers who had not churned.

Complaints and churn: Customers with recorded complaints had a 31.32% observed churn rate compared with 23.60% for customers without recorded complaints.

Plan-level differences: Observed churn rates were 24.21% for Basic, 26.60% for Standard, and 28.91% for Premium.

Subscription-type differences: Observed churn rates were 29.35% for Referral, 23.64% for Paid, and 25.58% for Organic.

Geographic variation: The analysis shows variation in both churn counts and observed churn rates across states.

Revenue impact: The analysis separates annualized revenue associated with active and churned customers, allowing churn to be considered from both customer-count and revenue perspectives.

11. Business Insights

- Understand the early customer lifecycle because shorter-tenure customers show substantially higher observed churn.
- Examine onboarding and early customer experience for new customers.
- Investigate complaint patterns and whether particular support issues are associated with higher churn.
- Monitor customer satisfaction and support interactions as part of retention analysis.
- Examine differences between subscription and plan types.

- Investigate geographic variation in churn.
- Consider revenue exposure when evaluating churn rather than looking only at the number of churned customers.

These observations describe patterns in the dataset and should be investigated further before being used for operational decisions.

12. Limitations

9. The dataset contains 521 customers, so the findings describe this dataset and should not automatically be generalized to a larger population.
10. Some support-related variables are available only for customers with recorded support interactions.
11. Observed relationships do not establish causation.
12. The analysis is descriptive and does not constitute a predictive churn model.
13. Some groups contain relatively small numbers of customers, so their observed churn rates should be interpreted carefully.
14. Revenue figures in the SQL section are annualized from monthly charges using $\text{monthly charges} \times 12$ and should be interpreted accordingly.

13. Tools and Technologies

- Microsoft Excel
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- MySQL
- SQL
- Power BI
- Jupyter Notebook

14. Conclusion

This project provides an end-to-end analysis of customer churn using Python, SQL, and Power BI.

The analysis began with raw customer, subscription, and support data. The datasets were cleaned and combined in Python, followed by feature engineering to create variables such as age groups, tenure groups, churn flags, annual charges, churn-risk groups, complaint counts, escalation counts, and customer satisfaction measures.

The prepared data was then loaded into MySQL and analyzed using SQL business questions. Finally, Power BI was used to create an interactive five-page dashboard covering overall churn, demographics, subscription and tenure, revenue and churn risk, and customer support.

The analysis identified substantial differences in observed churn across tenure groups, subscription types, plans, age groups, geographic areas, and complaint status.

Raw Data → Data Cleaning → Feature Engineering → Python Analysis → MySQL → SQL Analysis → Power BI Dashboard → Business Insights